

Eric Solak

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EDUCATION

McMaster University | Hamilton, ON

2022 - May 2026

Bachelor of Engineering, Software Engineering

cGPA: 3.70/4

WORK EXPERIENCE

AI Automation Consultant, Local SMB's | Toronto, ON

June 2026 - Present

- **Gathered requirements** and built AI agent workflows for local businesses to automate core operations: invoicing, document summarization, sales tracking, commission calculations, and pricing logic using Claude API.
- **Deployed end-to-end solutions in weeks, validated with customers** before handoff, demonstrating rapid iteration and customer-centric development.

CATTLElytics Inc. | Hamilton, ON

Sep 2025 - Apr 2026

Software Engineer

- **Led AI development** for Pocket AI, an AI-powered hands-free app for workers who can't stop to type: it listens to natural speech to create and manage tasks in real time. Delivered as a complete product to CATTLElytics, a platform managing **1 in 50 cows across North America**.
- **Directed a team of 5 engineers** through the full product development lifecycle in **8 months**, from requirements gathering and architecture design to production handoff, delivering a fully functional application now being integrated into CATTLElytics' platform.
- Integrated OpenAI Whisper for real-time audio transcription, validated through usability testing that achieved **98% system voice recognition accuracy** across 200 test commands under simulated barn noise, exceeding the 95% project target.
- Built the Flask backend powering voice-to-task workflows with REST APIs for authentication, task lifecycle management, with an **offline-first architecture**, ensuring reliable performance in low-connectivity farm environments.
- Designed and built mobile UI components in React Native (Expo), including task views, audio capture flows, and speech confirmation feedback; usability testing yielded a mean satisfaction score of 4.4/5.

PROJECTS

musicGPT · 45M-Parameter LLM from Scratch | Python, PyTorch

Jul 2026

- Built a **45M-parameter decoder-only transformer GPT entirely from scratch**, hand-implementing RoPE, causal self-attention, LayerNorm, and the training loop.
- Trained on a 0.92B-token pipeline with a hand-rolled markup stripper and 3x upweighting of music-domain content; **reduced val loss from 10.9 (uniform baseline) to 3.26** with no overfitting across 14,000 steps.
- Wrote a pytest suite for silent failure modes (attention equivalence, causality, RoPE geometry) that caught a **weight-tying logit leak** and a **vocab-padding crash** invisible to standard training metrics.

SneakPeek · AI Mobile Application | Python, PyTorch, React Native, JavaScript

Jan 2025 - Mar 2025

- Built a full-stack mobile app for **AI-powered sneaker identification** and social forum, where users upload images and receive AI-identified shoe model, brand, and purchase links.
- Trained a custom DenseNet201 AI model on a labelled dataset for sneaker identification, applying a data augmentation pipeline (perspective transforms, saturation jitter, geometric transforms) to maximize generalization.
- Architected a **multi-agent Blackboard system** where Google Gemini 2.0 Flash, GPT-4o, and the custom PyTorch model each submit predictions to a shared space; a central Gemini controller arbitrates and selects the final result.

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, HTML/CSS, SQL, C

Frameworks & Libraries: PyTorch, FastAPI, Flask, React Native, NumPy, Pandas, scikit-learn

Tools: Docker, AWS EC2, Render, Claude Code, GitHub Actions, JUnit, SQLite

INTERESTS

Self-taught musician (piano & guitar, 10 years) · chess · weightlifting · running